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PHANTOM CODE

Evading Windows 11 25H2 Through POSIX-Based Self-Deletion and Stealth Injection

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REALITY CHECK

The Post-Lunch Struggle

We know you're all fighting the food coma right now.

"If you fall asleep, we'll inject shellcode into your dreams."

SPEAKERS

Who Are We?

JV

Jakkaraju Varshith

Student
Rashtriya Raksha University

SPECIALIZATIONS

Offensive Security Researcher | Windows Exploitation | DFIR

BACKGROUND

- Team FORENSEEK – Cyber Security CTF & hackathon team
- Pursuing M.Sc. in Cybersecurity & Digital Forensics (Final Semester)

VJ

Vivek Joshi

Assistant Professor
Rashtriya Raksha University

SPECIALIZATIONS

OSINT | Windows Internals

BACKGROUND

- Assistant Professor, Rashtriya Raksha University
- Ph.D. Researcher in Machine Learning

CHAPTER 01

THE PROBLEM

Microsoft Broke Everything

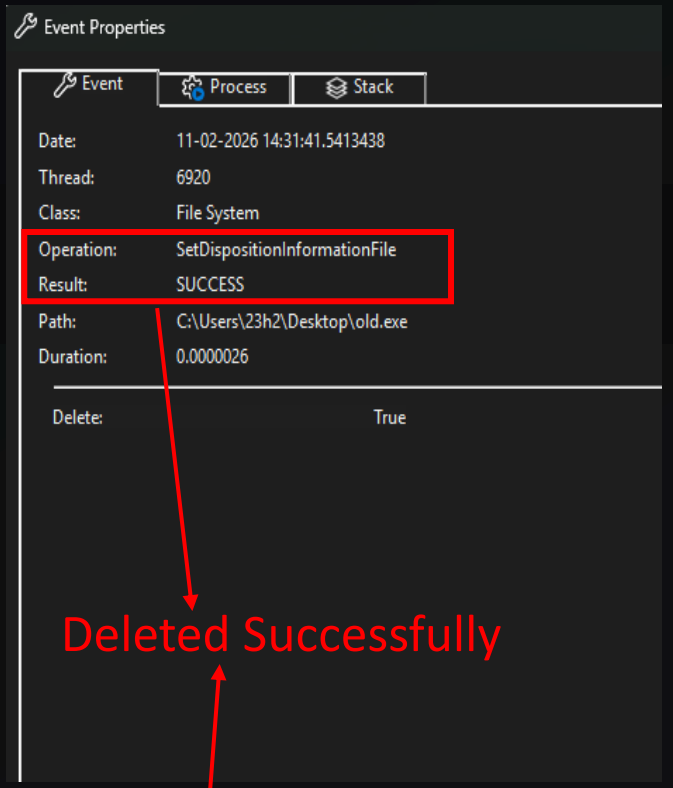
October 2024 – When Windows 11 24H2 (now 25H2) dropped and broke the self-deleting dropper overnight

ORIGINAL METHOD (2019)

The LloydLabs Technique

The method that worked beautifully... **until it didn't**:

1. **Open file with DELETE access**
   CreateFileW with DELETE permission
2. **Rename main :$DATA stream**
   SetFileInformationByHandle → :deadbeaf
3. **Close handle and reopen**
   Original filename, DELETE access
4. **Set delete disposition**
   FILE_DISPOSITION_INFO.DeleteFile = TRUE
5. **Close handle to trigger deletion**
   File appears deleted



The Old API

ADS

23H2

7276	QueryNameInformationFile	C:\Users\23h2\Desktop\old.exe	SUCCESS	Name: \Users\23h...
7276	CreateFile	C:\Users\23h2\Desktop\old.exe	SUCCESS	Desired Access: R...
7276	SetRenameInformationFile	C:\Users\23h2\Desktop\old.exe	SUCCESS	ReplaceIfExists: Fa...
7276	CloseFile	C:\Users\23h2\Desktop\old.exe:deadbeef	SUCCESS	
7276	CreateFile	C:\Users\23h2\Desktop\old.exe	SUCCESS	Desired Access: R...
7276	SetDispositionInformationFile	C:\Users\23h2\Desktop\old.exe	SUCCESS	Delete: True

THE INCIDENT

What Actually Happened?

October 2024

Microsoft released Windows 11 24H2 with undocumented API changes

The Impact

Most self-deletion techniques used by malware and red team tools suddenly stopped working

Windows PowerShell

```
PS C:\Users\25H2\Desktop\test_old_method> ls

    Directory: C:\Users\25H2\Desktop\test_old_method

Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         2/23/2026   1:43 AM        1089024 old_method.exe

PS C:\Users\25H2\Desktop\test_old_method> ls

    Directory: C:\Users\25H2\Desktop\test_old_method

Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         2/23/2026   1:43 AM              0 old_method.exe
```

Before Execution

Old method in 25H2

After Execution

"It works on my machine" – Most hackers after 24H2

ROOT CAUSE ANALYSIS

Why It Broke in 25H2

What Changed

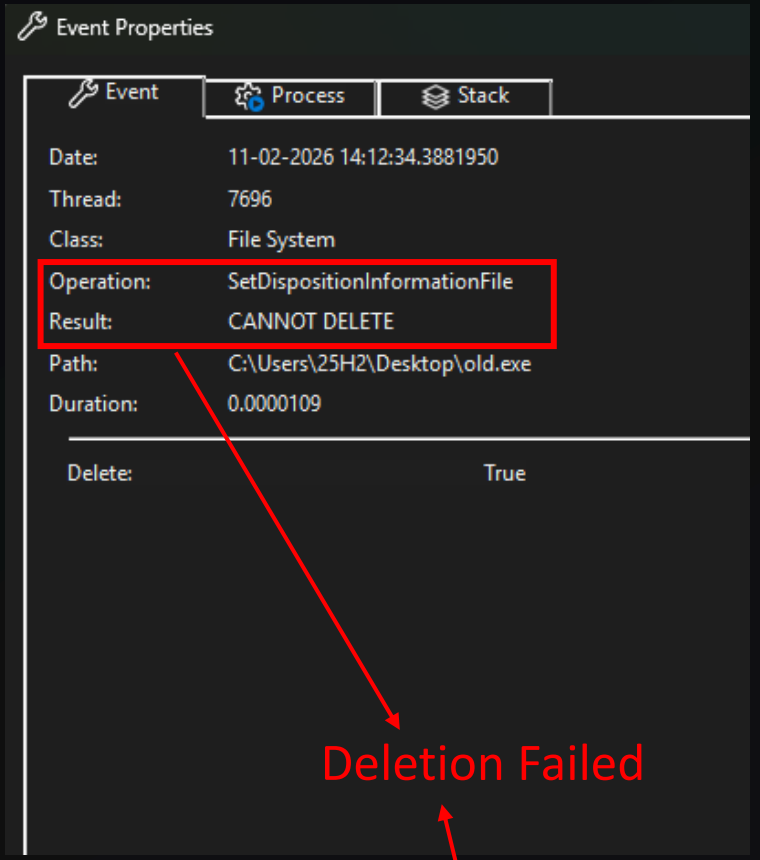
Microsoft's undocumented API modifications
Kernel-level changes to file system behavior

SetDispositionInfo function altered
Core deletion logic changed

The Failure Mode

SetDispositionInformationFile call fails
Returns STATUS_CANNOT_DELETE (0xc0000121)

ADS gets deleted but file remains as 0 bytes
Empty file – still a forensic artifact!



Deletion Failed

The same Old API

ADS

25H2

1936	QueryNameInformationFile	C:\Users\25H2\Desktop\old.exe	SUCCESS	Name: \Users\25H...
1936	CreateFile	C:\Users\25H2\Desktop\old.exe	SUCCESS	Desired Access: R...
1936	SetRenameInformationFile	C:\Users\25H2\Desktop\old.exe	SUCCESS	ReplaceIfExists: Fa...
1936	CloseFile	C:\Users\25H2\Desktop\old.exe:deadbeef	SUCCESS	
1936	CreateFile	C:\Users\25H2\Desktop\old.exe	SUCCESS	Desired Access: R...
1936	SetDispositionInformationFile	C:\Users\25H2\Desktop\old.exe	CANNOT DELETE	Delete: True
1936	CloseFile	C:\Users\25H2\Desktop\old.exe	SUCCESS	

THE CHALLENGE

The Research Question

The Challenge

How do we achieve **COMPLETE file deletion** on Windows 11 25H2?

Questions We Needed to Answer

- ? What mechanisms still work reliably?
- ? Can we find undocumented APIs or flags that bypass restrictions?
- ? How does WSL handle file deletions differently?
- ? LINUX USES UNLINK() — IMMEDIATE REMOVAL, EVEN WHILE A FILE IS OPEN. WINDOWS DOESN'T SUPPORT THIS... OR DOES IT?

CHAPTER 02

THE DISCOVERY

POSIX to the Rescue

How Windows Subsystem for Linux accidentally saved red teamers

THE BREAKTHROUGH

The WSL Connection

Key Insight

WSL needs POSIX-style file deletion semantics to work properly on Windows

The Discovery

FILE_DISPOSITION_POSIX_SEMANTICS flag
Value: 0x00000002

Originally designed for
Windows Subsystem for Linux compatibility

Unexpected use
Bypasses 25H2 NTFS restrictions entirely!

Flag Name	Value	Meaning
FILE_DISPOSITION_DO_NOT_DELETE	0x00000000	Specifies the system should not delete a file.
FILE_DISPOSITION_DELETE	0x00000001	Specifies the system should delete a file.
FILE_DISPOSITION_POSIX_SEMANTICS	0x00000002	Specifies the system should perform a POSIX-style delete. See more info in Remarks.
FILE_DISPOSITION_FORCE_IMAGE_SECTION_CHECK	0x00000004	Specifies the system should force an image section check.
FILE_DISPOSITION_ON_CLOSE	0x00000008	Specifies if the system sets or clears the on-close state.
FILE_DISPOSITION_IGNORE_READONLY_ATTRIBUTE	0x00000010	Allows read-only files to be deleted.

THE POSIX ADVANTAGE

Standard Windows Delete	× Blocks on open handles
POSIX Delete	✓ Unlinks immediately
Standard Cleanup	× leaves 0 byte file
POSIX Cleanup	✓ Complete removal

FILE_DISPOSITION_INFORMATION_EX structure (ntddk.h)

The **FILE_DISPOSITION_INFORMATION_EX** structure is used as an argument to the ZwSetInformationFile routine and indicates how the operating system should delete a file.

The Stealth Factor

Why This Is Serious:

During DFIR analysis, we discovered something **alarming** :

The process (25h2.exe) is NOT found in memory capture!

Minimal Artifacts Generated

✓ Prefetch (something executed)

✓ MFT (only prefetch entry visible)

✓ Registry (UserAssist from double-click)

✓ Sysmon (Process Create/Terminate only)

Almost invisible to forensic analysis!

DFIR ANALYSIS

Forensic Artifacts Analysis

Sysmon

Event Id	Process Id	Map Description	Payload Data1	Payload Data4	Executable Info
=	=	abc	abc	abc	abc
1	2052	Process creation	ProcessID: 8472…	ParentProcess: C:\Windows\explorer.exe	"C:\Users\25H2\Desktop\tools\25h2.exe"
1	2052	Process creation	ProcessID: 1768…	ParentProcess: C:\Users\25H2\Desktop\tools\25h2.exe	\??\C:\WINDOWS\system32\conhost.exe 0xffffffff -ForceV1
5	2052	Process terminated	ProcessID: 8472…		C:\Users\25H2\Desktop\tools\25h2.exe

Volatility.psxview

```
┌──(venv)(root㉿DHIA01T006)-[/mnt/e/tools/volatility3]
└─# python3 vol.py -f /mnt/e/personal/nullcon/artifacts/20260219.mem windows.malware.psxview.PsXView | grep -i 25h2.exe
Progress:  100.00               PDB scanning finished
┌──(venv)(root㉿DHIA01T006)-[/mnt/e/tools/volatility3]
└─#
```

Strings

```
┌──(venv)(root㉿DHIA01T006)-[/mnt/e/tools/volatility3]
└─# strings /mnt/e/personal/nullcon/artifacts/20260219.mem | grep -i 25h2.exe
25h2.exe
\Device\HarddiskVolume3\Users\c3ihub\Downloads\25h2.exe
9ChainStart,C:\Users\c3ihub\Downloads\25h2.exe,Genome,WinBlueRTM,Value,0a
25h2.exe
25h2.exe
C:\Users\c3ihub\Downloads\25h2.exe
"MonitorProcess,C:\Users\c3ihub\Downloads\25h2.exeT
25h2.exe
Y25h2.exe
^C
```

Prefetch and files loaded

Drag a column header here to group by that column

Line	Source Filename	Source Creat…	Source Modif…	Source Access…	Executable Name	Run Count
=	abc	=	=	=	abc	=
1	E:\personal\nullcon\artifacts\…	2026-02-20 0…	2026-02-16 1…	2026-02-20 04…	25H2.EXE	1

Cell contents

```
\VOLUME{01dc8967a00b5449-a6a01705}\WINDOWS\SYSTEM32\NTDLL.DLL,
\VOLUME{01dc8967a00b5449-a6a01705}\WINDOWS\SYSTEM32\C_1252.NLS,
\VOLUME{01dc8967a00b5449-a6a01705}\WINDOWS\SYSTEM32\C_437.NLS,
\VOLUME{01dc8967a00b5449-a6a01705}\WINDOWS\SYSTEM32\L_INTL.NLS,
\VOLUME{01dc8967a00b5449-a6a01705}\USERS\25H2\DESKTOP\TOOLS\25H2.EXE,
\VOLUME{01dc8967a00b5449-a6a01705}\$MFT,
\VOLUME{01dc8967a00b5449-a6a01705}\WINDOWS\SYSTEM32\KERNEL32.DLL,
\VOLUME{01dc8967a00b5449-a6a01705}\WINDOWS\SYSTEM32\KERNELBASE.DLL,
\VOLUME{01dc8967a00b5449-a6a01705}\WINDOWS\SYSTEM32\LOCALE.NLS,
\VOLUME{01dc8967a00b5449-a6a01705}\WINDOWS\SYSTEM32\DRIVERS\CONDRV.SYS,
\VOLUME{01dc8967a00b5449-a6a01705}\WINDOWS\SYSTEM32\CONHOST.EXE,
\VOLUME{01dc8967a00b5449-a6a01705}\WINDOWS\SYSTEM32\DRIVERS\APPID.SYS
```

Format Replace: CRLF CR LF Comma Tab Font size 12

$USN

Name	Exte…	Update Reasons	File Attributes
abc	abc	abc	abc
25h2.exe	.exe	FileCreate	Archive
25h2.exe	.exe	DataExtend|FileCreate	Archive
25h2.exe	.exe	DataExtend|FileCreate|BasicInfoChange	Archive
25h2.exe	.exe	DataExtend|FileCreate|BasicInfoChange|Close	Archive
25h2.exe	.exe	Close	Archive
25h2.exe	.exe	DataTruncation|StreamChange	Archive
25h2.exe	.exe	DataTruncation|StreamChange|Close	Archive
25h2.exe	.exe	RenameOldName	Archive
25H2.EXE-2D2179EF.pf	.pf	FileCreate	Archive|NotContentIndexed
25H2.EXE-2D2179EF.pf	.pf	DataExtend|FileCreate	Archive|NotContentIndexed
25H2.EXE-2D2179EF.pf	.pf	DataExtend|FileCreate|Close	Archive|NotContentIndexed

$MFT

Line	Entry Number	In Use	Parent Path	File Name	Is Directory	Has Ads	Is Ads	SI<FN	u Sec Zeros	Copied	Si Flags
=	abc	abc	abc	abc	abc	abc	abc	abc	abc	abc	abc
914	929	True	.\Windows\Prefetch	25H2.EXE-2D2179EF.pf	False	False	False	False	False	False	Archive|NotContentIndexed
51809	44099	True	.\PathUnknown\Directory with ID 0x0000040A-00000003\pre…	25H2.EXE-2D2179EF.pf	False	False	False	True	False	False	Archive

Forensic Artifacts Analysis (specific to 22h2)

192.168.171.138:49698 → 192.168.171.129:443
ESTABLISHED, orphaned socket (PID = -)

explorer.exe (PID 2004)
→ cmd.exe (PID 6872)
→ cmd.exe (PID 6912)

CmdLine Analysis
Both cmd.exe PIDs have blank cmdline

LdrModules Analysis
PIDs 6872, 6912, 6880
InLoad/InInit/InMem = False/False/False
PEB completely empty

File lifecycle:
11:05:58 FileCreate
11:06:24 Executed
11:06:29 DataTruncation + Rename
11:06:30 Process terminated

CHAPTER 03

DETECTION

What Blue Teams Need to Know

DETECTION VECTORS

Detection Opportunities

API Call Patterns

CreateToolhelp32Snapshot → VirtualAllocEx → CreateRemoteThread

Classic injection sequence

NtSetInformationFile with FileDispositionInformationEx

Unusual in normal applications

FILE_DISPOSITION_POSIX_SEMANTICS in non-WSL context

Red flag for non-Linux apps

Memory Indicators

⚠ RWX allocations in system processes

⚠ explorer.exe spawning network connections

KEY PRINCIPLE

Behavioral > Signature-based detection

Look for patterns, not just individual events

RESOURCES

Resources & Acknowledgments

GitHub Repository

https://github.com/0xOb5k-J/Self_Deleting_Exploit.git

✓ source code

✓ Technical documentation

References

- LloydLabs delete-self-poc (original technique)
- Microsoft FILE_DISPOSITION_POSIX_SEMANTICS docs

Contributors

- Mr. Pranjal Upadhyay

THANK YOU

Q&A